Information Systems

MIS Chapter 12 Highlights

Electronic meeting systems enable decision makers in different locations to participate in a group decision-making process.

Executive information systems (EISs) branches of DSSs, are interactive information systems that give executives easy access to internal and external data and typically include “drill-down” features and a digital dashboard for examining and analyzing information.

Semistructured decisions include a structured aspect that benefits from information retrieval, analytical models, and information systems technology.

Structured decisions or programmable tasks, can be automated because a well-defined standard operating procedure exists for these types of decisions.

Unstructured decisions are typically one-time decisions, with no standard operating procedure pertaining to them.

During the **choice phase**, the best and most effective course of action is chosen.

Decision support system (DSS) is an interactive information system consisting of hardware, software, data, and models (mathematical and statistical) designed to assist decision makers in an organization. Its three major components are a database, a model base, and a user interface.

In the **design phase**, the objective is to define criteria for the decision, generate alternatives for meeting the criteria, and define associations between the criteria and the alternatives.

A **digital dashboard** integrates information from multiple sources and presents it in a unified, understandable format, often as charts and graphs. It offers up-to-the minute snapshots of information and assists decision makers in identifying trends and potential problems.

A **geographic information system (GIS)** captures, stores, processes, and displays geographic information or information in a geographic context, such as showing the location of all city streetlights on a map.

Group support systems (GSSs) assist decision makers working in groups. These systems use computer and communication technologies to formulate, process, and implement a decision-making task and can be considered a kind of intervention technology that helps overcome the limitations of group interactions.

Groupware assists groups in communicating, collaborating, and coordinating their activities. It is a collection of applications that supports decision makers by providing access to a shared environment and information.

In the **implementation phase**, the organization devises a plan for carrying out the alternative selected in the choice phase and obtains the resources to implement the plan.

In the **intelligence phase**, a decision maker examines the organization's environment for conditions that need decisions. Data is collected from a variety of sources (internal and external) and processed. From this information, the decision maker can discover ways to approach the problem.

Management support systems (MSSs) are the different types of information systems that have been developed to support certain aspects and types of decisions. Each type of MSS is designed with unique goals and objectives.

A **managerial designer** defines the management issues in designing and using a DSS. These issues do not involve the technological aspects of the system; they are related to management's goals and needs.

The **model base component** includes mathematical and statistical models that, along with the database, enable a DSS to analyze information.

A **model builder** is the liaison between users and designers. He or she is responsible for supplying information on what the model does, what data inputs it accepts, how the model's output should be interpreted, and what assumptions go into creating and using the model.

The **technical designer** focuses on how the DSS is implemented and usually addresses questions about data storage, file structure, user access, response time, and security measures.